

Curriculum Vitae

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Profile

- Engineer with a solid background in software, data engineering, and machine learning
- Value-driven, versatile, and resilient professional
- Proven track record of delivering scalable solutions using agile methodologies
- Focus on automation, maintainability, best practices, and design patterns
- Thrives in dynamic environments with cross-functional teams
- Committed to applying and developing knowledge to solve real-world challenges in software, data, and machine learning domains

Main Skills

- **Programming Languages & Frameworks:** Python, PySpark, Java, JavaScript/Node.js/React/Angular, HTML/CSS/XML, R
- **Data & Analytics:** Databricks, Apache Airflow, Statistical Process Control, Machine Learning, Deep Learning
- **Databases:** SQL/RDBMS, Oracle, MongoDB, Cloudant, MS SQL Server, IBM DB2, Cassandra, PostgreSQL, DynamoDB
- **Cloud & DevOps:** Azure, AWS, Docker, Kubernetes, Helm, Azure DevOps, Travis CI, Terraform, AWS-CDK
- **Monitoring & Tools:** Datadog, Application Insights, OpenTelemetry
- **Other:** Linux/Bash Scripting, UML, Spring (Core/Boot/MVC/Data-JPA), JAX-WS & JAX-RS, JPA/Hibernate

Professional Experience

Data Engineer

Novo Nordisk, Søborg, Denmark

Nov 2023 – Present

- Designed, developed, and maintained GxP critical and non-critical data analytics solutions in the Safety Surveillance Data Science team
- Implemented data product solutions including project templates, persistence layer, data pipeline abstraction constructs, observability, data quality and data contracts

Technologies & Methodologies: AWS-CDK, AWS, Azure, Python, PySpark, Pandas, PyArrow, Databricks, Databricks Asset Bundles, Azure DevOps, Terraform, R, Bash.

Machine Learning Engineer

Pandora A/S, Copenhagen, Denmark
Oct 2020 – Nov 2023

- Designed, interfaced, and integrated data products in the Data Science & Machine Learning team
- Planned and provided technical steering as product owner
- Designed and implemented MLOps blueprints to support data science adoption and scaling in the organization

Technologies & Methodologies: Apache Airflow, Python, Apache PySpark, Databricks, Data Lake, Azure, Azure DevOps, Azure Functions, K8s, Terraform, Bash, SAFe, Java, Apache Beam.

Software Engineer

Digital, A.P. Moller - Maersk A/S, Copenhagen, Denmark
Feb 2019 – Oct 2020

- Implemented and maintained microservice APIs in the Future Pricing and Optimization team

Technologies & Methodologies: Java 8, Spring Boot, Hibernate, Swagger, Microsoft SQL Server, Azure, Docker, Scrum, TDD, Terraform, Kubernetes, Helm, Azure DevOps, Jenkins.

Software Engineer

8 West Consulting, Cork, Ireland
Nov 2017 – Feb 2019

- Implemented a data layer microservice API for Data Provider Accuracy

Technologies & Methodologies: Java 8, Spring Boot, Hibernate, Swagger, RAML, JPA, Oracle, Docker.

Software Engineer

IBM, Cork, Ireland
Jan 2014 – Nov 2017

IOA - Expert Advice - Implemented microservices for a cognitive application leveraging orchestration of machine learning services - Developed web UI, bot, and Slack application for the cognitive application

Technologies & Methodologies: Java, Dropwizard Framework, OpenNLP, Solr, Slack API, JavaScript, Node.js, Angular, HTML5, CSS, IBM Watson Retrieve & Rank, IBM Watson Natural Language Classifier, IBM Watson Conversation Service, Cloudant NoSQL DB, Git, IBM Bluemix, Cloud Foundry, IBM Bluemix DevOps Services.

Smart Cloud Analytics - Predictive Insights team - Developed and maintained software components in on-premise and cloud-based agile environments - Researched and conducted proof-of-concept experiments with various technologies and open-source libraries - Analyzed and assessed algorithm performance on real-time data sets

Technologies & Methodologies: Agile, Scrum, Java, WebSphere AS, Rational Team Concert, Design Patterns, Linux Bash Script, REST, Spring, MongoDB, Redis, Neo4j, DB2, Bluemix, R, Multithreading, Statistical Process Control, Node.js, Docker, Cloud Foundry.

Smart Cloud Application Performance Management team - Maintained and implemented user stories and bug fixes in Cloud and Smarter Infrastructure monitoring solutions

Technologies & Methodologies: Agile, Scrum, Java, WebSphere AS, Rational Team Concert, Python, Design Patterns, Linux Bash Script, Windows Scripting, REST, Spring.

Software Engineer

DeCare Systems, Cork, Ireland

Jul 2013 – Jan 2014

- Maintained and implemented bug fixes in a health care management web application
- Configured and programmed an ANSI 834 Health Care Enrollment (HIPAA compliant) middleware data interface

Technologies & Methodologies: Java, Struts, JavaScript, JSP, Hibernate, Oracle, Design Patterns; IBM WebSphere Transformation Extender (WTX), Oracle.

Software Engineer

Wincor Nixdorf, Lisbon, Portugal

Aug 2008 – Jul 2013

- Developed a retail web application

Technologies & Methodologies: Java, Spring, Spring MVC, jQuery, JPA/Hibernate, Derby, UML, Design Patterns.

- Designed and developed a fiscal library component for WN retail software solutions, ensuring legal compliance; included requirements analysis and certification support

Technologies & Methodologies: Linux, Windows, Java, C, C#, Bash, SQL, UML, PostgreSQL, MS SQL Server, Design Patterns, Test Driven Development, automated unit tests (JUnit) and integration test cases.

- Handled roll-out, support, and maintenance of retail software solutions for international customers (Portugal, Angola, Algeria); performed analysis, specification, configuration, maintenance, support, and training for standard and custom developments

Technologies & Methodologies: Linux, Java, C, Bash, Cocoon Framework (J2EE), UML, PostgreSQL, SQL, Design Patterns, Test Driven Development, automated unit tests (JUnit) and integration test cases.

- Executed customer specifications for customizing and rolling out a mobile & automated checkout solution in a Portuguese retailer; developed online data synchronization scripts

Technologies & Methodologies: UML, Java, Linux, Windows, Derby DB, Bash and Windows Scripting.

Software Engineer

EID, S.A., Almada, Portugal

Jan 2007 – Aug 2008

- Implemented a GUI Management Console for setting up Routing Tables in an Organization MMHS infrastructure

Technologies & Methodologies: Java, Swing API, UML, Design Patterns, RUP, Test Driven Development, automated unit tests (JUnit), pair programming, and integration test cases.

Software Consultant

AXA Insurances, Lisbon, Portugal

Jan 2006 – Jan 2007

- Developed call-centre script-guide software
- Performed maintenance on organizational process solutions for desktop/rich-client and web applications

Technologies & Methodologies: VB6, COM, C#, SQL, UML, XML, MS Exchange CDO & MAPI, ADO & ADO.Net, Log4Net, NUnit, NAnt, Office Automation Objects, Oracle 9i, MS Access, MS SQL Server, Design Patterns, Test Driven Development.

Consulting Engineer

Care4IT, Lisbon, Portugal

Oct 2004 – Jan 2006

- Analyzed, designed, and implemented a health care management solution for a specific pathology care process
- Analyzed, designed, and implemented a custom software solution for pathology care process monitoring and clinical guideline adherence
- Conducted consulting analysis for an integrated stock management solution in healthcare; assessed technologies and suppliers; mapped processes and performed workflow analysis

Technologies & Methodologies: UML, Java, Swing API, MS SQL Server, Design Patterns, Test Driven Development, automated unit testing (JUnit); UML, VB6, MS Access.

Quality Engineer Internship

Sival, Leiria, Portugal

Apr 2004 – Oct 2004

- Implemented Statistical Process Control on PVC tubes in a short-run production process, applying 4 types of control charts including multivariate analysis (final year project)

Educational Qualifications

Higher Diploma in Data Science and Analytics

Cork Institute of Technology

Feb 2017 – Dec 2018

Subjects: Statistics & Mathematics, Computer Science and Data Science, Time Series, Multivariate Analysis, Data Mining & Knowledge Discovery, Data Visualization, Machine Learning, Deep Learning.

Post-Graduate Studies in Lean Management

NOVA School of Science and Technology, NOVA University Lisbon

Oct 2008 – Jul 2009

Subjects: Lean Thinking, Value Stream Mapping, TPM, SMED, JIT Systems, Six-Sigma, Kaizen Events, Lean Supply Chain Management.

Industrial Engineering University Degree

NOVA School of Science and Technology, NOVA University Lisbon

Sep 1996 – Dec 2004

Subjects: Production Management, Industrial Logistics, Quality Engineering, Ergonomics, IT, Industrial Automation, Computer Aided Production and Manufacturing Technology.

References

Excellent references available on request.